



IF3170 Inteligensi Artifisial

Institut Teknologi Bandung

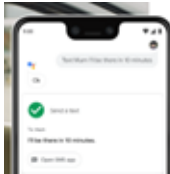
What is AI

Semester I TA 2026/2027

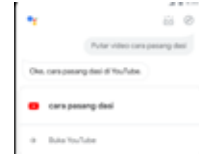




AI In Life: AI Assistant



“Hey Google, text
Mum I'll be there in 10
minutes”



“Hey Google, play my
morning playlist”



“Hey Google, dim the
bedroom lights”

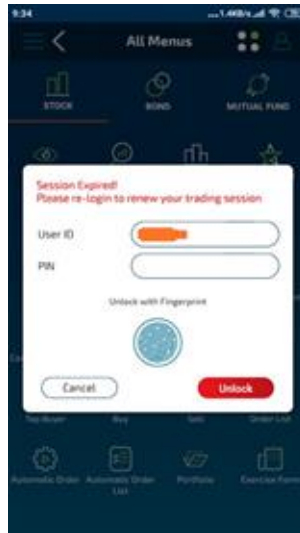


“Hey Google, set the
temperature to 20
degrees”



AI In Life: Biometric Verification

Fingerprint verification



Face verification



<https://www.grab.com/sg/driver-2/selfieverification/>



AI In Life: OpenAI for Dota 2



OpenAI Five

Competitive: 7.215–42 (99.4% winrate, 15.019 total players)

Note: During the live stream, the game count incorrectly omitted games abandoned by the human side.

Cooperative: 35.466 games (18.689 total players)

Source: <https://v.yimg.com/v/yh-Usa06l8X8/maze/default.jpg>

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www.itb.ac.id



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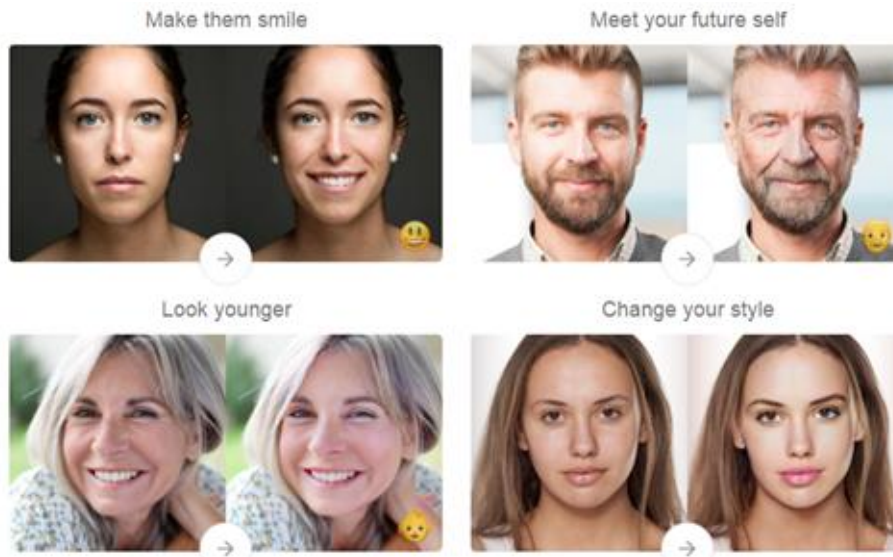
AI In Life: AI Alphastar StarCraft II



<https://techcrunch.com/2019/01/24/starcraft-ii-playing-ai-alphastar-takes-out-pros-and-defeats/>
https://www.youtube.com/watch?v=PFMRDm_H9Sg



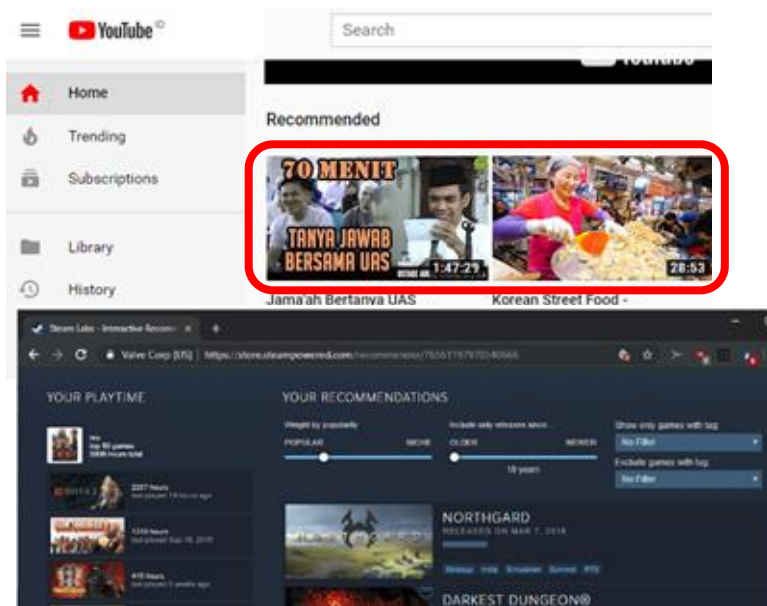
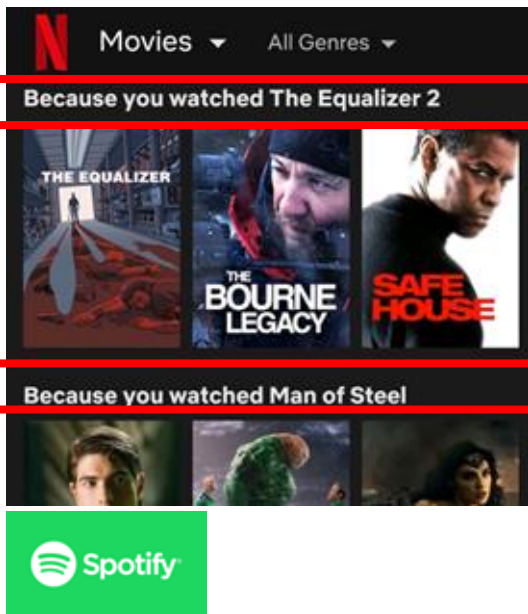
AI In Life: Transform Your Face



<https://www.faceapp.com/>

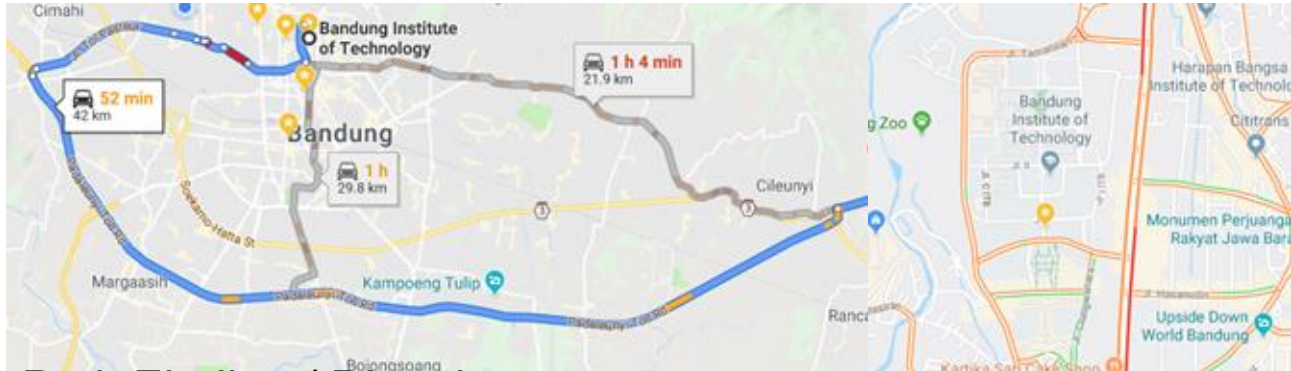


AI In Life: Recommender System





AI In Life: AI in Traffic

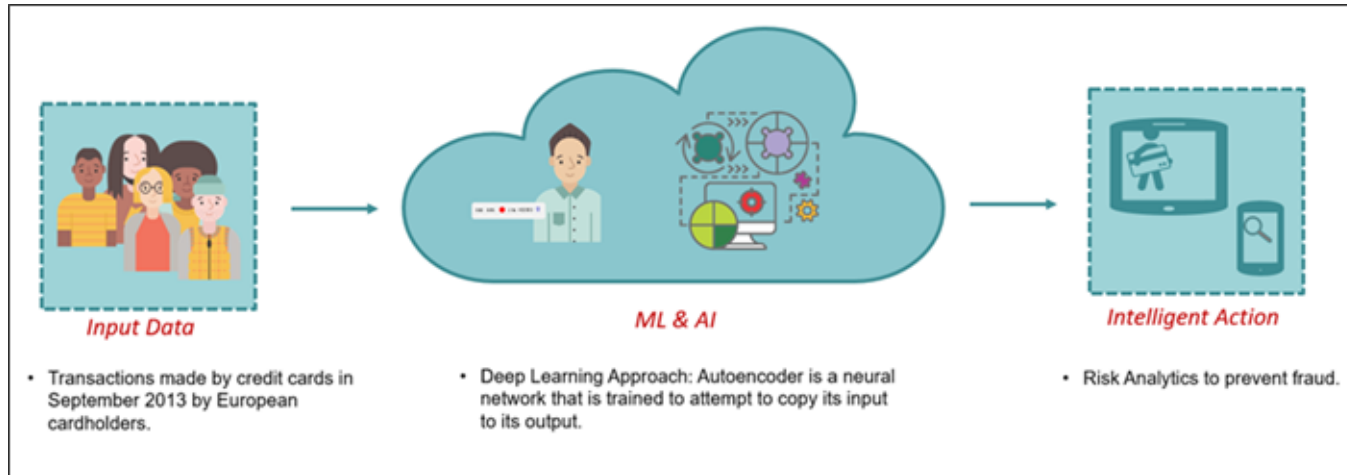


Path Finding / Direction

Traffic pattern



AI In Life: Fraud Detection



Source: https://miro.medium.com/max/3848/1*3R2cLLJ349O5G6V5rZJXxw.png



Kiva: Amazon Warehouse Robot



Source: https://image.cdn.com/api/v1/image/33487775-An_employee_jacks.jpg?v=512907015&w=1400&h=950



AI for Health Care

10 AI Applications That Could Change Health Care

APPLICATION	POTENTIAL ANNUAL VALUE BY 2026	KEY DRIVERS FOR ADOPTION
Robot-assisted surgery	\$40B	Technological advances in robotic solutions for more types of surgery
Virtual nursing assistants	20	Increasing pressure caused by medical labor shortage
Administrative workflow	18	Easier integration with existing technology infrastructure
Fraud detection	17	Need to address increasingly complex service and payment fraud attempts
Dosage error reduction	16	Prevalence of medical errors, which leads to tangible penalties
Connected machines	14	Proliferation of connected machines/devices
Clinical trial participation	13	Patent cliff; plethora of data; outcomes-driven approach
Preliminary diagnosis	5	Interoperability/data architecture to enhance accuracy
Automated image diagnosis	3	Storage capacity; greater trust in AI technology
Cybersecurity	2	Increase in breaches; pressure to protect health data

SOURCE: ACCENTURE

© HBR.ORG

<https://hbr.org/2018/05/10-promising-ai-applications-in-health-care>

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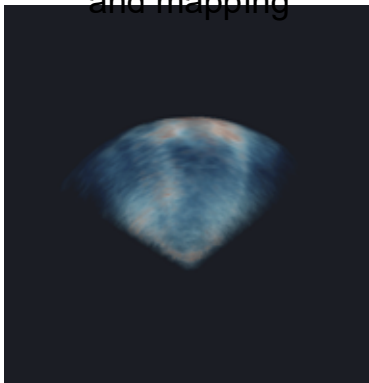


AI in Medical Image Analysis

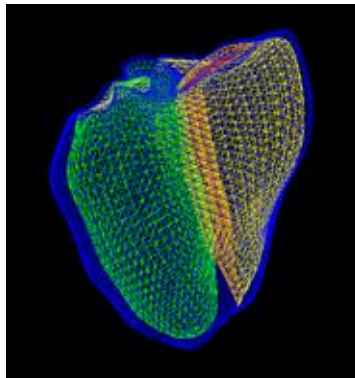
Pressure
quantification



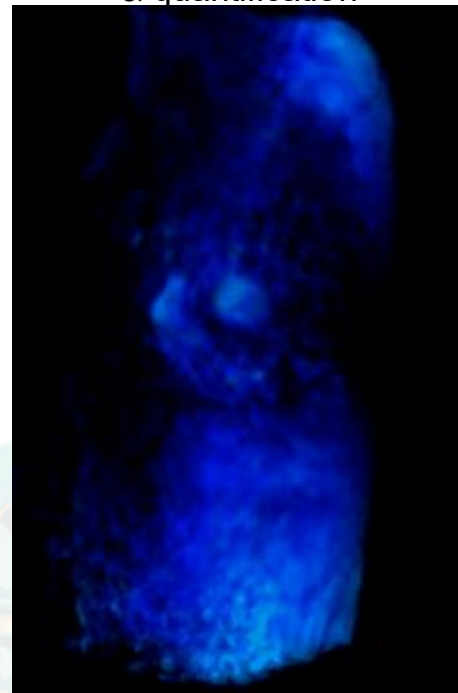
3D reconstruction
and mapping



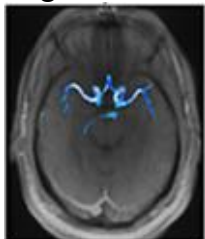
Surface mapping



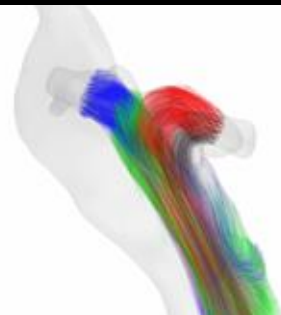
Flow visualization
& quantification



Segmentation



Geometry
reconstruction



Flow simulation

Courtesy of E. Ferdian & A. Elsayed (UoA), D. Marlevi (MIT), and D. Zhao (ABI)





AI for Manufacturing



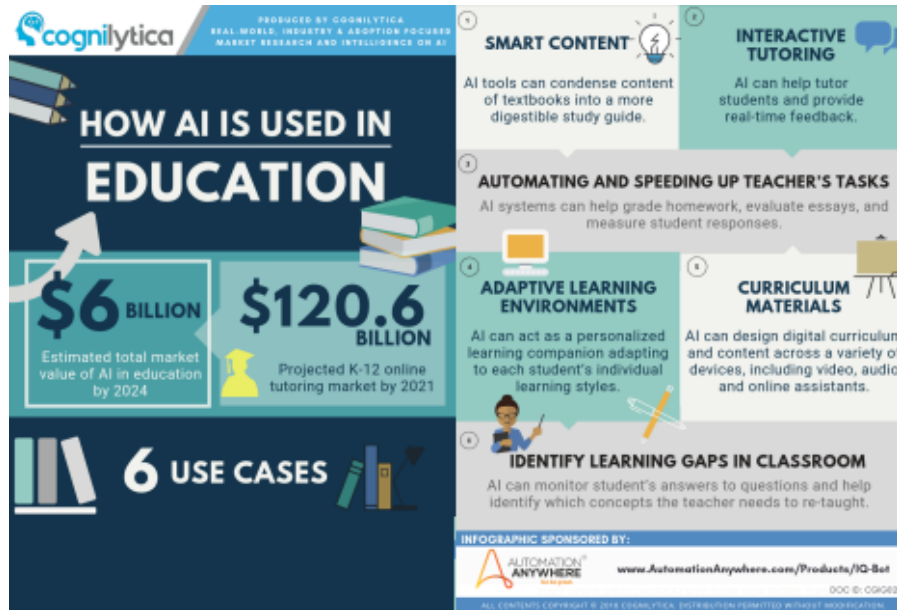
Q: How relevant are the following concepts for your company?
Base: all respondents

<https://www.forbes.com/sites/louis.columbus/2018/03/11/10-ways-machine-learning-is-revolutionizing-manufacturing-in-2018/#61e28f2523ac>

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AI for Education



<https://www.cognilytica.com/2019/03/28/infographic-ai-in-education/>

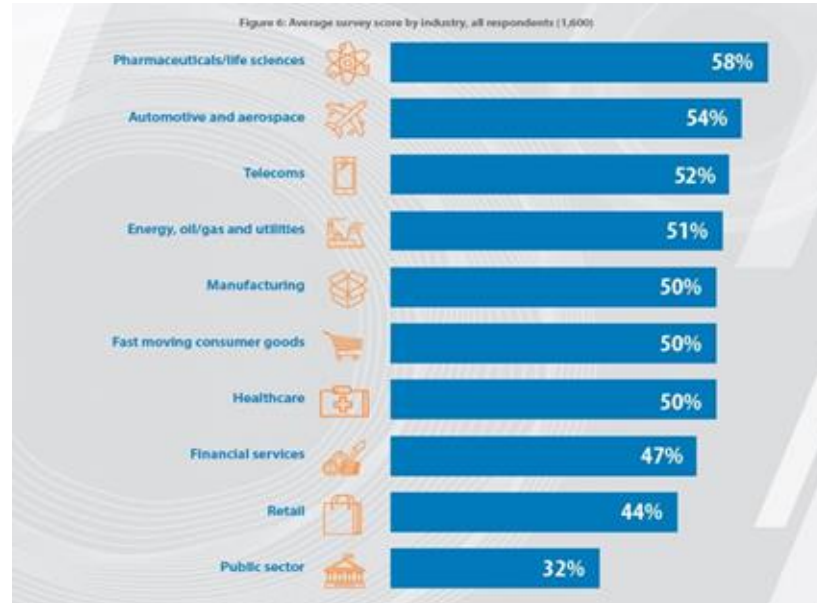
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AI Adoption by Industry

Percentage of market players who plan to adopt AI in next two years across various business verticals (Infosys Survey, 2018)

<https://existek.com/blog/ai-programming-and-ai-programming-languages/>





AI In Life: Self Driving Car



HOW WAYMO'S SELF-DRIVING CAR WORKS

One of Waymo's three lidar systems that shoots lasers so the car can see its surroundings. Waymo says this lidar can detect a helmet two-football fields away.

A forward facing camera works with 8 others stationed around the car to provide 360 degrees of vision.

Radar sensors can detect objects in rain, fog, or snow.

Waymo's self-driving sensors are tightly integrated into the hybrid minivan created by Fiat Chrysler.



SOURCE: Waymo

BUSINESS INSIDER

[adar.com/news/uber-self-driving-cars](http://www.adar.com/news/uber-self-driving-cars)

Google Waymo (8M miles): early rider program.
(public trial at Phoenix, Arizona)

<http://www.businessinsider.sg/how-does-googles-waymo-self-driving-car-work-graphic-2017-1/?r=US&IR=T>



Self Driving Truck First



Otto (Uber) self-driving truck delivered freight in Arizona

<https://www.recode.net/2018/3/6/17082774/uber-self-driving-truck-otto-freight-embark-waymo-alphabet-arizona> (6 Mar 2018)

<https://www.techemergence.com/self-driving-trucks-timelines/> (Juli 2018)

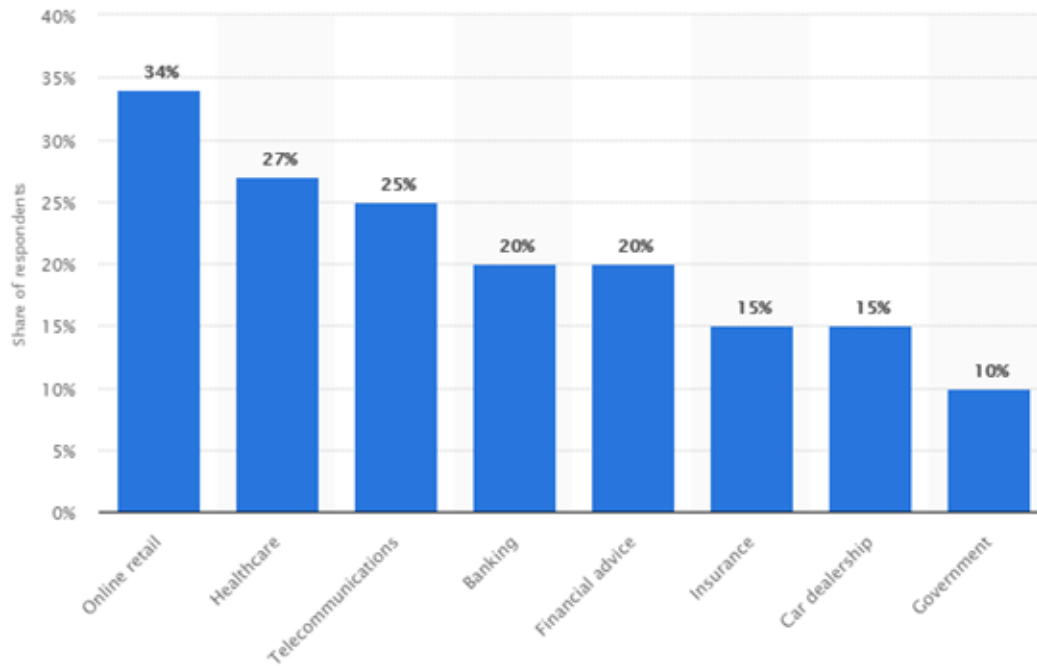
<https://www.trucks.com/2016/03/21/daimler-tests-self-driving-truck-platoon-in-live-traffic/> (21 Maret 2016)



Daimler Tests Self-Driving Truck Platoon in Live Traffic



AI in Life: Chatbots



Acceptance of artificial intelligence chatbots by customers worldwide, as of 2017, by service



AI in Life: Vacuuming Robot



Vacuuming Robot



Mopping Robot

<http://www.irobot.com/>





AI Introduction

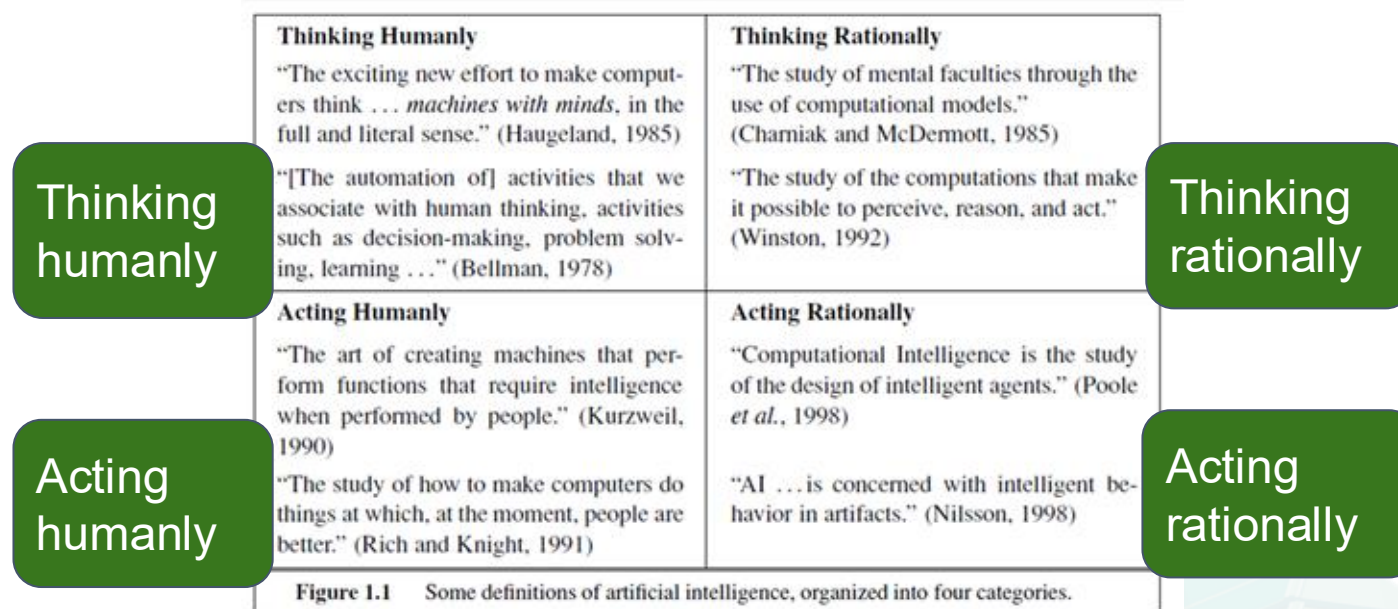
What is AI ?

AI applications





So.. What is AI? 8 Definitions, 4 Approaches





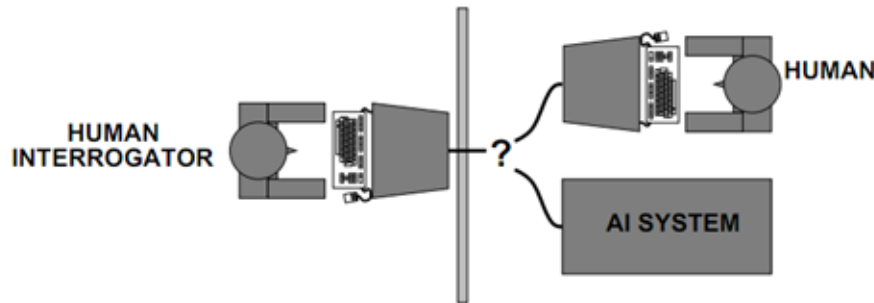
What is Artificial Intelligence ?

- The art of creating machines that perform functions that require intelligence when performed by people
(Kurzweil, 1990)
- The study of how to make computers do things at which, at the moment, people are better
(Rich and Knight, 1991)
- **AI: acting humanly**





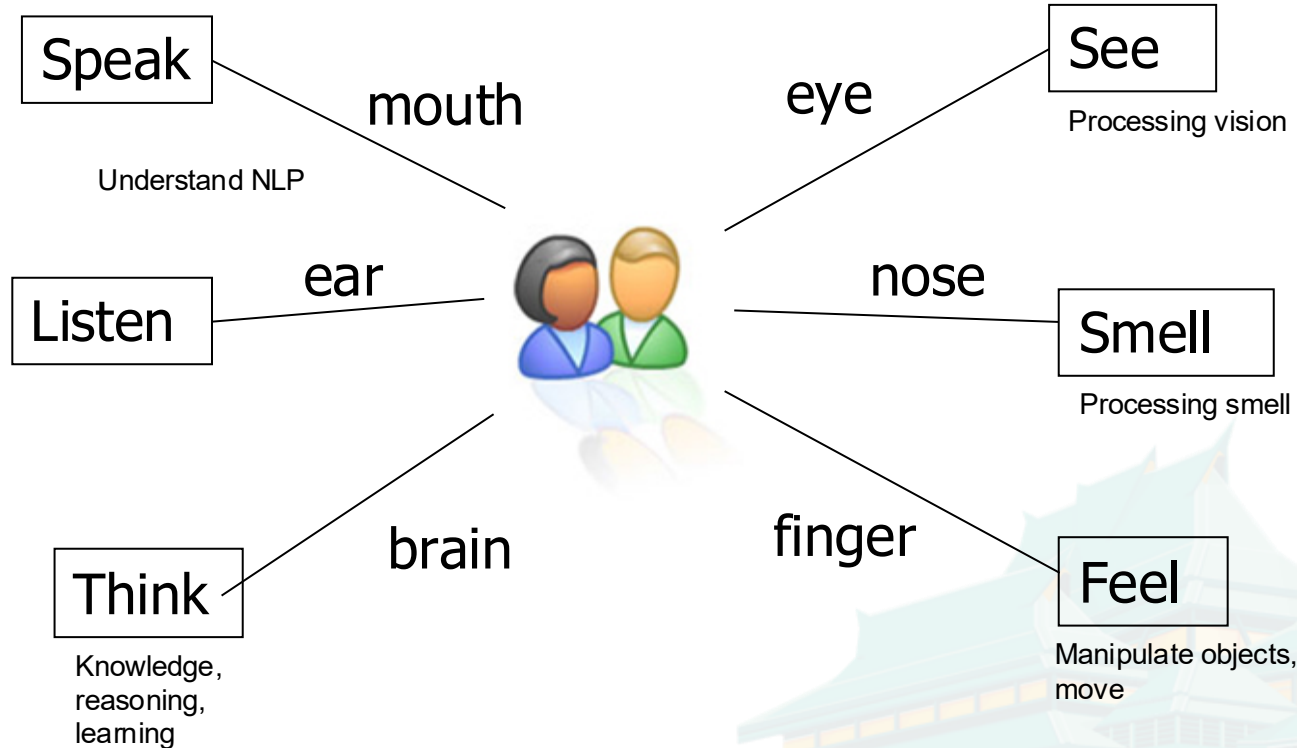
Acting humanly: Turing Test Approach



- Turing Test approach
 - A satisfactory operational definition of intelligence
 - Pass: human interrogator cannot tell whether the responses come from a person or from a computer

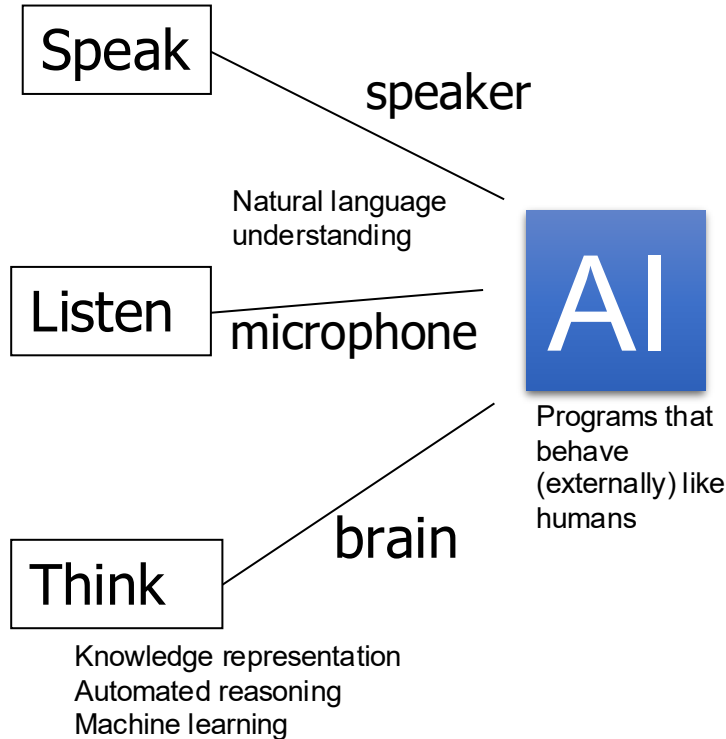


Human Capabilities





AI Capabilities for Turing Test





Loebner Prize

- Contest in Artificial Intelligence (AI) to find the world's best conversational chatbot computer program
- ended quietly and became defunct around 2020



Loebner 2013 winner :
<http://www.mitsuku.com/>

Loebner 2014 winner:
Rose by Bruce Wilcox
(developed for 2,5 years)

Loebner 2015 winner:
Pandorabot - Mitsuku
<http://www.pandorabots.com/pandora/talk?botid=933ae25bfe35dad2>

Loebner 2016 winner: Mitsuku

Loebner 2017 winner : Mitsuku

Loebner 2018 Winner: Mitsuku

Loebner 2019 Winner: Mitsuku

PC Therapist III, 1991

PC Professor - discusses Men
versus Women, 1992

PC Politican - discusses Liberals
versus Conservatives, 1993

...



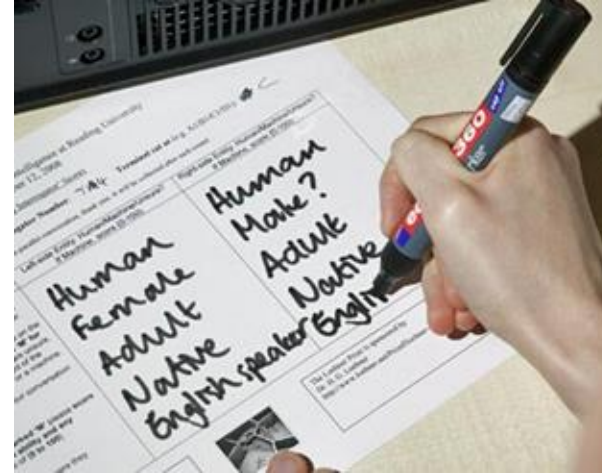
Meet Kuki!

Mitsuku, or Kuki to her close friends, is a record-breaking, five-time winner of the Loebner Prize Turing Test and the world's best conversational chatbot (according to folks like Google AI Research). As featured in the New York Times, Wall Street Journal, BBC, Guardian, Wired, and Radiolab.





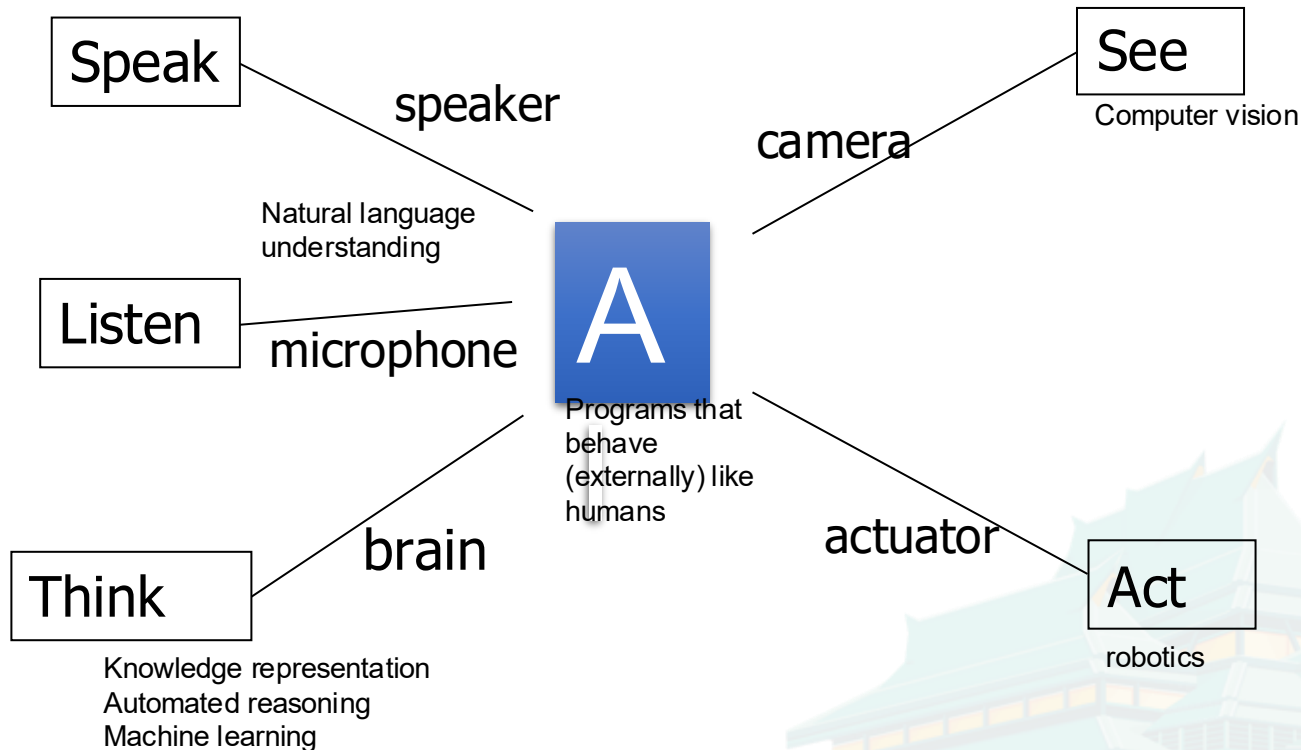
Loebner Prize Turing Test Competition



<http://www.computerweekly.com/feature/Meet-Elbot-Loebner-Prize-Turing-Test-contest-winner-transcript-and-photos>



AI Capabilities for Total Turing Test





AI: Acting humanly

- It is more important to study the underlying principles of intelligence than to duplicate it.
 - Little effort to pass the Turing Test





What is Artificial Intelligence ? (2)

- The exciting new effort to make computers think ... machines with minds, in the full and literal sense
(Haugeland, 1985)
- [The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ...
(Bellman, 1978)
- **AI: thinking humanly**





What is AI: Thinking humanly

- Computational models of human “thought” processes? (**Thinking humanly**)
 - Programs that operate (internally) the way humans do
 - How humans think ? : actual working of human minds
 - Introspection: trying to catch our own thoughts as they go by
 - Psychological experiment: observing a person in action
 - Brain imaging: observing the brain in action
 - Cognitive modeling approach
 - Computer models from AI and experimental techniques from psychology to construct precise and testable theories of human mind
 - Machines with minds



What is Artificial Intelligence ? (3)

- The study of mental faculties through the use of computational models
(Charniak and McDermott, 1985)
- The study of the computations that make it possible to perceive, reason, and act
(Winston, 1992)
- **AI: thinking rationally**





What is AI: Thinking rationally

- Computational systems that **behave** intelligently? (**Thinking rationally**)
 - What does it mean to behave intelligently?
 - Laws of thought approach
 - Right thinking: reasoning





What is Artificial Intelligence ? (4)

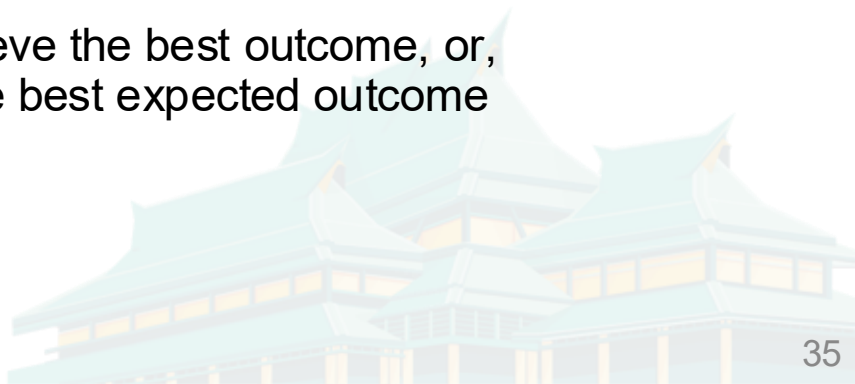
- Computational intelligence is the study of the design of intelligent agents (Poole et al., 1998)
- AI...is concerned with intelligent behavior in artifacts (Nilsson, 1998)
- **AI: acting rationally**





What is AI: Acting rationally

- Computational systems that behave rationally!
(Acting rationally)
 - Goal, knowledge : act rationally
 - Correct inference is not all of rationality; in some situations, there is no provably correct thing to do, but something must still be done.
 - Later: rational agent
 - Agent that acts so as to achieve the best outcome, or, when there is uncertainty, the best expected outcome





AI Revolution is On

- Today's AI doesn't try to re-create the brain.
 - we still don't really understand how the brain works, much less how to re-create it.
- It uses machine learning, massive data sets, sophisticated sensors, and clever algorithms to master discrete tasks.

http://www.wired.com/magazine/2010/12/ff_ai_essay_airevolution/



AI vs not AI Applications

- AI Applications: problems in computer science that don't feel well specified enough for the rest of the computer science community to want to work on.
- Those problems could evolve out of AI.
- **By definition, no AI ever works; if it works, it's not AI.**
 - Information retrieval, data mining, game technology, etc ?
- But there are all kinds of applications of AI.
- AI applications are very viable.





Automation vs AI

Automation	AI
Make the system able to complete work (tasks) in the same way every time	Make the system capable of responding to changes in information/needs from the surrounding environment dynamically; and provide interpretive Decisions
Suitable for tasks that do not require changes over time	Suitable for relatively complex tasks that require making interpretive decisions
Potential to: increase productivity, reduce operational costs, simplify workflow	Potential to: increase productivity, lower operational costs, create intelligent workflows that can respond to dynamically changing needs



References

- [1] A lecturer from MIT, “Techniques in Artificial Intelligence”
based on Russell’s book





Exercise

1. Pengenalan gambar dengan menggunakan hierarchical temporal memory yang meniru prinsip biologi dari neocortex merupakan contoh pengembangan ilmu AI dengan pendekatan...
2. Dengan berbekal pengetahuan (algoritma) dan tujuan (goal) yang ingin dicapai, sistem mencari rangkaian aksi untuk menghasilkan solusi terbaik, merupakan definisi AI dengan pendekatan ...
3. Pembuatan sistem pakar yang mencoba mengimitasi perilaku pakar dalam mendiagnosis kemungkinan penyakit tanaman berdasar pengamatan terhadap perilaku pakar tersebut merupakan contoh teknologi AI dgn pendekatan..
4. Sebuah persoalan logic puzzle, diselesaikan dengan pendekatan pencarian memanfaatkan algoritma backtracking, termasuk dalam sistem berbasis AI dengan pendekatan ...
5. Persoalan teka-teki yang diselesaikan dengan menggunakan kaidah-kaidah inferensi logika (modus ponens, modus tollens, dll) termasuk system AI dengan pendekatan ...



Exercise: AI vs non-AI

1. Software akuntansi yang menggunakan berbagai rumus penghitungan rugi laba dengan input berupa file xls dengan struktur kolomnya sudah pasti dan nilai sel berupa angka.
2. Sistem lampu lalu lintas yang rentang waktu menyalanya lampu merah/kuning/hijau ditentukan berdasar jumlah kendaraan yang lewat dari arah tertentu, dihitung secara otomatis berdasar gambar yang tertangkap kamera cctv di atas lampu lalu lintas tsb.
3. Sistem tanya jawab pemesanan kamar hotel menggunakan flowchart yang sudah ditentukan di dalam program. Pelanggan memberikan input dengan memilih dari kandidat input yang disediakan program.
4. Perangkat lampu yang menyala jika ada objek yang berjalan di bawah lampu tersebut. Lampu akan padam jika sudah tidak ada objek bergerak di bawah lampu untuk satu waktu tertentu



Thank You

